

Mohd Waqar

Software Engineer — Full-Stack & AI Applications

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Poonch, J&K, India | Remote, GMT+5:30 (flexible EU/US overlap)

SUMMARY

Full-stack engineer (B.Tech '26) building production software and AI systems end-to-end. RAG assistant raised eval pass rate from 38% to 74% and cut hallucinations from 38% to 8% on a 100-query labeled eval, serving 1,000+ daily queries with cited answers; sole developer of a live NZ property SaaS (oakandkey.co.nz). 3 internships across fintech, web, and AI/ML. Strongest in TypeScript, Node.js, React, Python, PostgreSQL; specialty in LLM engineering (RAG, pgvector, multi-model routing). Remote-ready: async, self-directed delivery.

TECHNICAL SKILLS

AI/LLM: RAG (pgvector, hybrid BM25 + dense search, reciprocal rank fusion, cross-encoder reranking), LLM evaluation (labeled eval sets, LLM-as-judge), multi-model routing (OpenAI GPT-4o, Claude, xAI Grok), prompt engineering, MCP, semantic caching, guardrails
Languages: TypeScript, JavaScript (ES6+), Python, SQL **Core CS:** Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks

Frontend: React, Next.js, Redux, Tailwind CSS, Responsive Design, Accessibility (WCAG)

Backend: Node.js, Express, REST APIs, WebSockets, PostgreSQL, Supabase, Redis, JWT, OAuth 2.0 **Cloud/DevOps:** AWS (EC2, S3, Lambda), Docker, GitHub Actions (CI/CD), Vercel, k6, rate limiting

Tools: Jest, Vitest, Postman, Git, Agile/Scrum

EXPERIENCE

Software Development Engineer Intern

Jul 2025 – Aug 2025

Bluestock Fintech (fintech education startup)

Remote

- Rebuilt 3 React dashboard modules (replacing legacy jQuery) for 20+ daily ops users, cutting average task-completion time by 15%.
- Raised API test coverage from 30% to 75% with Jest across 12 critical endpoints, integrated into CI/CD; production bug recurrence fell 20% over 6 weeks.

Web Development Intern

Feb 2025 – Jun 2025

International Data Solution

Remote

- Delivered 3 production React + Tailwind CSS web apps for external e-commerce and healthcare clients, on-time across all sprints.
- Built a 20+ component reusable UI library with PR review standards, cutting new-developer onboarding from 5 days to 2.

AI/ML Development Intern

Jun 2024 – Aug 2024

Next24Tech

Remote

- Benchmarked 4 classifiers (Random Forest, XGBoost, SVM, LSTM) on financial fraud data, reaching 85% accuracy (12 pts above baseline); automated Pandas + scikit-learn preprocessing for 5 datasets (50K–200K rows), cutting manual effort about 30%.

PROJECTS

Oak & Key — Production Property Management SaaS

Live: oakandkey.co.nz

- Sole developer: designed, built, and shipped end-to-end for a New Zealand property firm (React, TypeScript, Supabase, PostgreSQL) — lease tracking, rent collection, bond lodgements, maintenance workflows, and interactive financial dashboards (Recharts).
- Built data-ingestion pipelines parsing Excel (xlsx) and PDF (pdf-lib) forms, enabling bulk imports of inspections, tenant applications, and payment ledgers.

Yojana Dost — RAG-Powered Knowledge Assistant

yojana-dost.vercel.app

React, Node.js, PostgreSQL (pgvector), OpenAI GPT-4o, Claude, Grok github.com/waqarbytes/Yojana-dost

- Built production RAG pipeline (pgvector embeddings, hybrid dense + BM25 retrieval with reciprocal rank fusion, cross-encoder reranking) over 150+ scheme documents; on a 100-query labeled eval harness, pass rate rose from 38% to 74%, hallucination rate fell from 38% to 8%, faithfulness rose from 2.1 to 4.6/5, with 86% citation precision and 82% recall@4.
- Cut p95 latency from 294 ms to 16 ms on cache hits (18x) and LLM cost per query by 53% via semantic caching (0.92 cosine threshold); multi-model routing with automatic fallback across OpenAI, Anthropic, and xAI APIs; live /api/metrics telemetry; 50-VU k6 load with 0% errors behind sliding-window rate limiting; 100% refusal on out-of-scope probes.
- Beta-tested with 50+ users at 1,000+ daily queries; instrumented per-query cost, latency, and failure modes to drive iteration.

MCP Server — India Schemes Toolkit (Model Context Protocol) github.com/waqarbytes/india-schemes-mcp

- Built and published an MCP server (TypeScript, Zod) exposing 6 authenticated tools — scheme search, eligibility checking, deadlines, comparisons — with schema validation and rate limiting; usable from Claude Desktop and any MCP-compatible client.

EDUCATION

B.Tech, Computer Engineering Aug 2022 – May 2026
Baba Ghulam Shah Badshah University, Poonch, J&K, India

CERTIFICATIONS

Azure AI & Cloud Fundamentals — IBM (Expected Mar 2026) | **AWS Cloud Practitioner Essentials** (course) — AWS, Jul 2025 | **Meta Front-End Developer Certificate** — Coursera/Meta, Mar 2025